■ Developer Mock Test Results

Test ID:	abb79a00-55cf-44...	**Student ID:**	N/A
Test Type:	Developer Assessment	**Date:**	2026-03-04 16:58:12
Overall Score:	10/25 (40.0%)	**Performance:**	■ Average
Warnings:	0/3	**Status:**	Completed ■

■ Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	3/10	30.0%	■■ Needs Work
■ MCQ/Theory	3/10	30.0%	■■ Needs Work
■ Coding	4/5	80.0%	■ Pass

■ Detailed Question Review

■ APTITUDE SECTION (3/10 - 30.0%)

■ Q1. If it is true that all students are learners and some learners are teachers, what can be concluded?

Your Answer: Some students are teachers
Correct Answer: Some learners are students
■ *The correct answer is "Some learners are students" because we know that all students are learners, which implies that the set of students is a subset of the set of learners. The user's answer, "Some students are teachers", is incorrect because while it may be true, it is not a direct conclusion from the given statements. The user likely made a mistake by trying to relate students directly to teachers, rather than considering the relationship between students and learners.*

■ Q2. If it is true that all dogs are animals and some animals are cats, what can be concluded?

Your Answer: Some dogs are cats
Correct Answer: Some animals are dogs
■ *The correct answer is "Some animals are dogs" because we know that all dogs are animals, so it logically follows that some animals (at least the ones that are dogs) are indeed dogs. The user's answer, "Some dogs are cats", is incorrect because the statement "some animals are cats" does not imply that those animals are also dogs. The user likely made a mistake by trying to combine the two statements in a way that isn't logically supported, rather than recognizing that the statement about dogs being animals directly implies that some animals are dogs.*

■ Q3. A mixture of juice and water is in the ratio 3:5. If 3 cups of juice are added to the mixture, what is the new ratio of juice to water?

Your Answer: 6:5
■ *Excellent! Right answer.*

■ Q4. What is the next number: 1, 3, 5, 7, 9, ?

Your Answer: 13
Correct Answer: 11
■ *The correct answer is 11 because the pattern of the sequence is adding 2 to the previous number, which is a classic example of an arithmetic sequence with a common difference of 2. The key calculation step is: 9 + 2 = 11. The user's likely mistake was adding 4 instead of 2, resulting in 13, which breaks the established pattern of the sequence.*

■ Q5. If it is true that all A are B and some B are not C, what can be concluded about A and C?

Your Answer: No A are C
Correct Answer: Some A are C
■ *The correct answer is "Some A are C" because we know all A are B, and while some B are not C, it doesn't exclude the possibility that some B (and therefore some A) are indeed C. The user's likely mistake was assuming that since some B are not C, none of the A (which are a subset of B) can be C, which is a logical fallacy. This incorrect assumption led them to conclude "No A are C" instead of recognizing the possibility of overlap between A and C.*

■ Q6. If 18 workers can build a house in 12 days, how many workers will be needed to build the same house in 10 days?

Your Answer: 22 workers
■ *Correct! Well done.*

■ Q7. If 4 men can do a job in 10 days, how many days will it take for 2 men to do the same job?

Your Answer: 18
Correct Answer: 20
■ *To find the number of days it will take for 2 men to do the job, we can use the formula: (number of men originally) * (number of days originally) = (number of men now) * (number of days now). So, 4 men * 10 days = 2 men * x days, which gives us 40 = 2x, and then x = 40 / 2 = 20. The user's answer of 18 is likely incorrect because they may have made a calculation error, such as incorrectly dividing the total work by the new number of men.*

■ Q8. A recipe calls for a ratio of 2:3 of flour to sugar. If you use 4 cups of flour, how many cups of sugar will you need?

Your Answer: 6 cups
■ *Excellent! Right answer.*

■ Q9. A mixture of paint and water is in the ratio 2:3. If 2 cups of paint are added to the mixture, what is the new ratio of paint to water?

Your Answer: 3:2
Correct Answer: 4:3
■ *The correct answer is 4:3 because the original ratio of paint to water is 2:3, and when 2 cups of paint are added, the amount of paint increases while the amount of water remains the same. The key calculation step is: let's assume the original amount of paint is 2x and water is 3x, then 2 cups of paint are added, making the new amount of paint 2x + 2, and since x can be chosen such that 2x = 2 (i.e., x = 1 for simplicity), the new ratio becomes (2 + 2):3 = 4:3. The user's likely mistake was not properly accounting for the increase in paint relative to the fixed amount of water.*

■ Q10. A store buys a product for $30 and sells it for $25. What is the loss percentage?

Your Answer: 15%
Correct Answer: 16.67%
■ *To find the loss percentage, we need to calculate the loss amount and divide it by the cost price, then multiply by 100. The key calculation step is: Loss percentage = ((Cost price - Selling price) / Cost price) * 100 = ((30 - 25) / 30) * 100 = (5 / 30) * 100 = 16.67%. The user's answer of 15% is likely incorrect because they may have made a calculation error or rounded incorrectly.*

■ MCQ SECTION (3/10 - 30.0%)

■ Q1. What is an array in Java?

Your Answer: A collection of elements of different data types

Correct Answer: A collection of elements of the same data type

■ *The correct answer is A) A collection of elements of the same data type, because in Java, an array is a data structure that stores a fixed-size collection of elements of the same data type. The user's answer, B) A collection of elements of different data types, is incorrect because Java arrays are homogeneous, meaning all elements must be of the same type. The user's likely mistake is a misunderstanding of the definition of an array in Java, which is a fundamental concept in programming that requires all elements to be of the same data type.*

■ Q2. What is synchronization in Java?

Your Answer: The process of coordinating access to non-shared resources

Correct Answer: The process of coordinating access to shared resources

■ *The correct answer, "The process of coordinating access to shared resources", is right because synchronization in Java is a mechanism that allows only one thread to access a shared resource at a time, preventing data inconsistencies and thread interference. This ensures that multiple threads cannot modify the same resource simultaneously, maintaining data integrity. The user's answer is likely incorrect because non-shared resources do not require synchronization, as only one thread can access them at a time by default.*

■ Q3. What happens when a Java file is compiled?

Your Answer:

```
It is converted to a .java file
```

Correct Answer:

```
It is converted to a .class file
```

■ *The correct answer is A) It is converted to a .class file, because when a Java file is compiled, the Java compiler translates the .java file into an intermediate format called bytecode, which is stored in a .class file. This .class file can then be executed by the Java Virtual Machine (JVM). The user's answer, B) It is converted to a .java file, is likely incorrect because .java files are the source code files that are compiled, not the result of the compilation process.*

■ Q4. What is method overriding in Java?

Your Answer:

```
A subclass provides the same implementation of a method
```

Correct Answer:

```
A subclass provides a different implementation of a method
```

■ *The correct answer is A) A subclass provides a different implementation of a method, because method overriding in Java allows a subclass to provide a specific implementation of a method that is already defined in its superclass. This enables the subclass to specialize the behavior of the method to suit its own needs. The user's answer, B) A subclass provides the same implementation of a method, is likely incorrect because the purpose of method overriding is to provide a different implementation, not to duplicate the same one.*

■ Q5. What are the types of data that can be stored in a variable in Java?

Your Answer: Int, double, char, String, arrays

■ *Correct! Good understanding.*

■ Q6. What is optimization in Java?

Your Answer: The process of improving the security of a program

Correct Answer: The process of improving the performance of a program

■ *The correct answer is "The process of improving the performance of a program" because optimization in Java refers to the process of modifying a program to make it more efficient, scalable, and faster, which ultimately improves its performance. The user's answer, "The process of improving the security of a program", is incorrect because while security is an important aspect of programming, optimization specifically focuses on enhancing the program's execution speed, memory usage, and overall efficiency. The user likely mistook optimization for a broader concept that encompasses all aspects of program improvement, including security, rather than focusing on performance.*

■ Q7. What is the base class for all exceptions in Java?

Your Answer: Throwable

■ *Correct! Well done.*

■ Q8. What are control structures in Java?

Your Answer: Methods and classes
Correct Answer: Conditions and loops
■ *The correct answer, "Conditions and loops", is right because control structures in Java refer to the elements that control the flow of a program's execution, such as if-else statements (conditions) and for/while loops. The user's answer, "Methods and classes", is incorrect because these are fundamental components of Java programming, but they do not specifically control the flow of a program. The user likely mistook the building blocks of Java programs for the control structures that dictate their execution.*

■ Q9. What is required to run Java programs?

Your Answer: Java JRE
Correct Answer: Java JDK
■ *The correct answer is Java JDK because it includes the Java Runtime Environment (JRE) and development tools, such as the Java compiler, which are necessary to run Java programs. The user's answer, Java JRE, is incorrect because while it can run Java programs, it does not include the development tools needed to compile them. The user likely mistook the JRE, which is a subset of the JDK, as the primary requirement to run Java programs, when in fact the JDK is the more comprehensive package that includes the JRE.*

■ Q10. How can you compile and run a Java program?

Your Answer: Using a code editor or IDE
■ *Correct! Good understanding.*

■ CODING SECTION (4/5 - 80.0%)

■ Q1. Write a Java program that reads a string from stdin and checks if it is a valid IP address. **Input Format:** - Line 1: A string **Output Format:** ...

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        if (!sc.hasNextLine()) {
            System.out.println("Not Valid IP Address");
            return;
        }

        String ip = sc.nextLine().trim();
        sc.close();

        String[] parts = ip.split("\\.");

        // Check if IP has exactly 4 parts
        if (parts.length != 4) {
            System.out.println("Not Valid IP Address");
            return;
        }

        for (String part : parts) {
            // Check if part is empty or contains non-digit characters
            if (part.isEmpty() || !part.matches("\\d+")) {
                System.out.println("Not Valid IP Address");
                return;
```

```
            }

            int num = Integer.parseInt(part);

            // Check if number is between 0 and 255
            if (num < 0 || num > 255) {
                System.out.println("Not Valid IP Address");
                return;
            }

            // Optional: Prevent leading zeros like 01, 001
            if (part.length() > 1 && part.startsWith("0")) {
                System.out.println("Not Valid IP Address");
                return;
            }
        }

        System.out.println("Valid IP Address");
    }
}
```

■ *Your code has successfully passed all test cases, indicating a correct and effective implementation of the problem solution. Great job on achieving an "Accepted" status, keep up the good work!*

■ Q2. Write a Java program that reads a matrix of integers from stdin and finds the maximum element. **Input Format: - Line 1: The number of rows in the ...**

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int rows = sc.nextInt();
        int cols = sc.nextInt();

        int max = Integer.MIN_VALUE;

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < cols; j++) {
                int num = sc.nextInt();
                if (num > max) {
                    max = num;
                }
            }
        }

        sc.close();
        System.out.println(max);
    }
}
```

■ *Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and effective implementation. Great job on achieving an "Accepted" status, indicating that your solution meets all requirements.*

■ Q3. Write a Java program that reads a string from stdin and decrypts it using a simple substitution cipher. **Input Format: - Line 1: A string **Outpu...**

Your Answer:

```
import java.util.Scanner;
```

```
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String input = sc.nextLine();
        sc.close();

        StringBuilder decrypted = new StringBuilder();

        for (char ch : input.toCharArray()) {
            if (Character.isLetter(ch)) {
                char base = Character.isUpperCase(ch) ? 'A' : 'a';
                char decryptedChar = (char) ((ch - base - 3 + 26) % 26 + base);
                decrypted.append(decryptedChar);
            } else {
                decrypted.append(ch); // keep non-letters unchanged
            }
        }

        System.out.println(decrypted.toString());
    }
}
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        String decrypted = decrypt(input);
        System.out.println(decrypted);
        scanner.close();
    }

    public static String decrypt(String input) {
        StringBuilder decrypted = new StringBuilder();
        for (char c : input.toCharArray()) {
            if (c >= 'a' && c <= 'z') {
                decrypted.append((char) ((c - 'a' - 3 + 26) % 26 + 'a'));
            } else {
                decrypted.append(c);
            }
        }
        return decrypted.toString();
    }
}
```

■ *The student's code is applying a Caesar cipher decryption with a shift of 3, but in the wrong direction. The expected output 'you' from input 'ruoy' suggests a simple reversal of the string, not a Caesar cipher decryption. The actual output 'orlv' indicates that the student's decryption logic is incorrect for this specific problem.*

■ Q4. Write a program that reads two integers and prints their least common multiple (LCM). **Input Format:** - Line 1: First integer (integer) - Line 2: S...

Your Answer:

```
import java.util.Scanner;

public class Main {

    // Method to calculate GCD using Euclidean algorithm
    public static int gcd(int a, int b) {
```

```
        while (b != 0) {
            int temp = b;
            b = a % b;
            a = temp;
        }
        return a;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int num1 = sc.nextInt();
        int num2 = sc.nextInt();
        sc.close();

        int gcdValue = gcd(num1, num2);

        int lcm = Math.abs(num1 * num2) / gcdValue;

        System.out.println(lcm);
    }
}
```

■ *Your code has successfully passed all test cases, indicating a correct and effective implementation of the solution. Great job on achieving an "Accepted" status, keep up the good work!*

■ Q5. Write a Java program that reads a string from stdin and encrypts it using a Caesar cipher. **Input Format:** - Line 1: A string - Line 2: The shift v...

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String input = sc.nextLine();
        int shift = sc.nextInt();
        sc.close();

        StringBuilder encrypted = new StringBuilder();

        // Normalize shift in case it is greater than 26
        shift = shift % 26;

        for (char ch : input.toCharArray()) {
            if (Character.isLetter(ch)) {
                char base = Character.isUpperCase(ch) ? 'A' : 'a';
                char encryptedChar = (char) ((ch - base + shift) % 26 + base);
                encrypted.append(encryptedChar);
            } else {
                encrypted.append(ch); // keep spaces and special characters unchanged
            }
        }

        System.out.println(encrypted.toString());
    }
}
```

■ *Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and its solution. Great job on achieving an "Accepted" status, indicating that your code is correct and efficient.*

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
